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5من الدقائق

ما هو Controller Factory في ASP.NET Core MVC؟

أثناء تعلمي لـ ASP.NET Core MVC، اتعرفت على... المزيد

Controller Factory in ASP.NET Core MVC

How MVC creates controllers and why a factory is used



1 What is a Controller Factory?

A Controller Factory is a component in ASP.NET Core MVC responsible for **creating controller instances** when a request comes to the application. Instead of creating controllers manually using **new**, MVC uses a factory to create them and resolve their dependencies from the **Built-in Dependency Injection (DI) container**.

2 Why Do We Need It?

- **Supports Dependency Injection**
  Controllers can receive services through constructor.
- **Centralized Creation Logic**
  Handles how controllers are created and released.
- **Allows Customization**
  We can replace the default factory with our own logic.
- **Improves Maintainability**
  Keeps the controller creation process consistent.

3 How It Works?

HTTP Request
A request comes to the application

→ **MVC**
MVC determines the correct controller

→ **Controller Factory**
Creates the controller using DI container

→ **Controller**
Controller instance is created with its dependencies

→ **Action Execution**
The requested action method is called

4 Default Implementation

ASP.NET Core uses a built-in implementation called **DefaultControllerFactory**.

```
public class DefaultControllerFactory : IControllerFactory
{
    public object CreateController(ControllerContext context)
    {
        var controllerType = context.ActionDescriptor
            .ControllerTypeInfo.AsType();
        // Create controller using DI
        return context:HttpContext.RequestServices
            .GetRequiredService(controllerType);
    }

    public void ReleaseController(ControllerContext context,
            object controller)
        // Called when the request is finished
    }
}
```

5 Can We Customize It?

Yes! We can create our own custom controller factory by implementing the **IControllerFactory** interface and register it in the DI container.

```
public class CustomControllerFactory : IControllerFactory
{
    // Custom logic to create controllers
}
builder.Services.AddSingleton<IControllerFactory,
CustomControllerFactory>();
```

6 Conclusion

Controller Factory is a key part of ASP.NET Core MVC that creates controllers, supports dependency injection, and allows customization, making the application more flexible and maintainable.

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2من الأيام

Excited to share that I've completed C# Advanced Design from DotNet With Elwakeel with a score of 39/40 (98%)!... المزيد

عرض الترجمة



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